

MST-Technologies and Processes

August, 16th, 2022

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task
- write with a pen NOT with a pencil
- if you hinder the exam you will be removed from the lecture hall and fail the exam (5.0 with the statement "Störung")

Cheating

- any attempt of cheating results in immediate failure of the exam (5.0 with the statement "Täuschung")
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
 - in case you feel unwell during the exam, you may withdraw during the exam for medical reasons; you need to provide (within 3 days) a written note from your doctor confirming the withdrawal for medical reasons
 - please see:
<https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams>
 - you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>
-

Necessary Points: **50 of 100 in total**

Points: _____

Result: _____

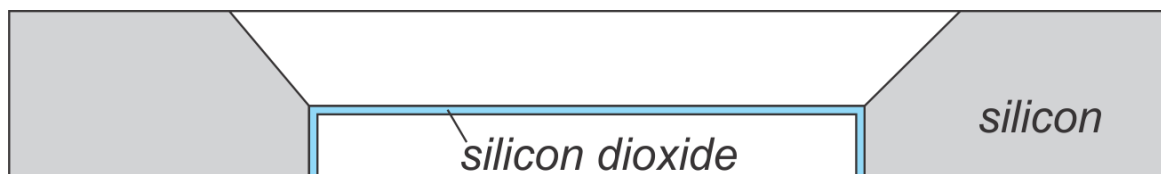


1. Processes

25

Sketch the manufacturing steps required to fabricate the following structure. The device is a microfluidic heat exchanger which transfers heat from a heating liquid (upper channel) to a process liquid (lower channel). Please note that the channels need to be closed - however, this is NOT part of the construction task. The channel walls of the lower channel need to be protected with a silicon dioxide layer as the process liquid degrades pure silicon.

Sketch and briefly explain each step. Each manufacturing process (e.g., lithography) used should be depicted out entirely when you first use it. If you use the same process again, you do not need to redraw the entire process.



2. Cleanroom Technology

25

2.1 Which of the following statements on working in the Cleanroom is correct (simply tick the box)? 10

- ☐ Wearing cleanroom trousers is a must in a cleanroom
- ☐ Wearing a hood or at least a hair cover is a must in a cleanroom
- ☐ Wearing a face mask is a must in a cleanroom
- ☐ Wearing a cleanroom overall is a must in a cleanroom
- ☐ If garment got wet during work, change it immediately
- ☐ Wearing boots or protective cover shoes is a must in a cleanroom
- ☐ Protective clothing in the cleanroom protects us and the substrates we work on
- ☐ Avoid skin contact with the process media, substrates and process items
- ☐ Wearing safety goggles is a must in a cleanroom
- ☐ Do not apply makeup or powder under your garment
- ☐ Suit up bottom-to-top: first boots, then trousers, then coat, then mask, then hood
- ☐ Wearing gloves is a must in a cleanroom
- ☐ Do not do not wear rings and wristwatches as they can damage the gloves or garment
- ☐ Do not work in the cleanroom when you suffer from cold or eczema

2.2 The following image shows a very common type of cleanroom setup:



Which type of cleanroom is this?

3

Name two distinct advantages of this setup.

2

2.5 Filter-fan units (FFUs) make use of four distinct filtering effects. Sketch and briefly explain how each of these effects works.

10

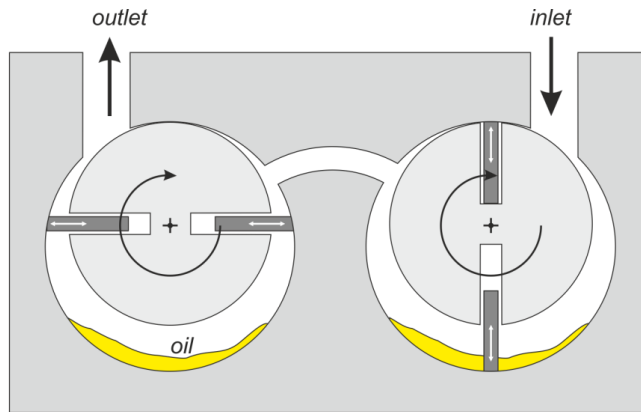


3. Vacuum technology

18

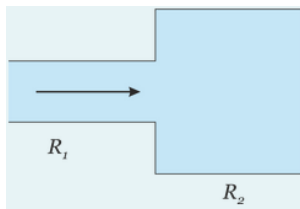
3.1 Which type of vacuum pump is depicted here?

2



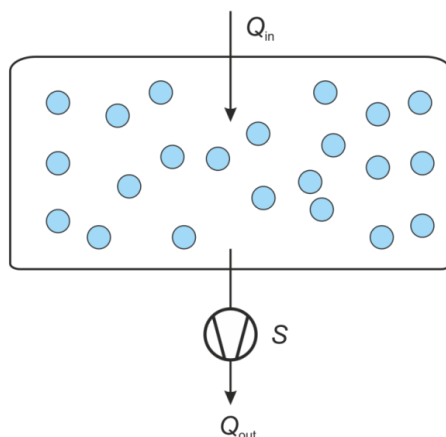
3.2 A hydraulic element with the resistance $R_1 = 9.5 \text{ s/m}^3$ is connected in series to a second hydraulic element with the resistance $R_2 = 5.5 \text{ s/m}^3$. What is the overall hydraulic resistance R_{tot} of the system?

4



3.3 The following schematic depicts a vacuum chambers which is to be kept at $p = 0.05 \text{ mbar}$. The pumping speed of the vacuum pump is $S = 200 \text{ l/s}$. What is the maximum allowed gas load Q_{in} which the pump can cope with if the pressure is to be sustained?

4



3.4 Sketch and briefly explain the turbomolecular pump.

5

Why does the turbomolecular pump always require a backing pump?

3



4. Thermal oxidation of silicon

12

4.1 In the Deal-Grove model, the following variables are used:

4

- D as the diffusivity of the oxidant in silicon dioxide
- k as the reaction rate of the oxidation of silicon to silicon dioxide
- N as the number of oxidant molecules required to produce one unit volume of oxide
- C as the ambient concentration of the oxidant
- $A = \frac{2D}{k}$ as a simplification factor
- $B = \frac{2DC}{N}$ as a second simplification factor

State the equation for the layer thickness $x(t)$ at higher layer thicknesses, i.e., for $x(t > t_c)$.

4.2 For usage in the Deal-Grove model we determined the following factors:

4

- $A = 1 \mu\text{m}$
- $B = 0.25 \mu\text{m}^2/\text{h}$

What is the characteristic time t_c ? [You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]

4.3 Using the same parameters as above, we now run an oxidation process for 4 hours (which is longer than t_c). What is the expected layer thickness we achieve? [You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]

4



5. MEMS Materials

20

5.1 The following are short explanation of typical defects found in MEMS materials. Which defects are these explanations referring to?

4

1. occurs when a single plane is shifted
2. disruption in the perfect lattice structure as one plane is missing
3. boundaries of near-order in polycrystalline materials
4. disruption in the perfect lattice structure as one additional plane is added

5.2 Sketch and explain the Czochralski Process for the generation of single-crystalline semiconductor-grade silicon

10

5.3 Indicate the following planes in a unit cell

6

